CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 9: Matrices

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Multiplication of matrices

1. If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, find A^2. (a) $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix}$ (c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ICSE 2025, 1 mark

SOLUTION

$$A^2 = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0+1 & 0+0 \\ 0+0 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Answer: (c)

2. Given $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ and $C = \begin{bmatrix} -5 & 1 \\ 7 & -4 \end{bmatrix}$, find (a) $A + C$ (b) $B(A + C)$ (c) $5B$ (d) $B(A + C) - 5B$. ICSE 2025, 4 marks

SOLUTION

(a) $A + C = \begin{bmatrix} 1-5 & 2+1 \\ 3+7 & 4-4 \end{bmatrix} = \begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix}$

(b) $B(A + C) = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}\begin{bmatrix} -4 & 3 \\ 10 & 0 \end{bmatrix} = \begin{bmatrix} -8+10 & 6+0 \\ -16+20 & 12+0 \end{bmatrix} = \begin{bmatrix} 2 & 6 \\ 4 & 12 \end{bmatrix}$

(c) $5B = \begin{bmatrix} 10 & 5 \\ 20 & 10 \end{bmatrix}$

(d) $B(A + C) - 5B = \begin{bmatrix} 2-10 & 6-5 \\ 4-20 & 12-10 \end{bmatrix}$

Answer: (d) $\begin{bmatrix} -8 & 1 \\ -16 & 2 \end{bmatrix}$

3. If $A = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}$ and $A^2 = \begin{bmatrix} 4 & x \\ 0 & 4 \end{bmatrix}$, find x. (a) 2 (b) 4 (c) 8 (d) 10 ICSE 2024, 1 mark

SOLUTION

$$A^2 = \begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & 2 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 4+0 & 4+4 \\ 0+0 & 0+4 \end{bmatrix} = \begin{bmatrix} 4 & 8 \\ 0 & 4 \end{bmatrix}$$

Answer: (c)

4. Given $A = \begin{bmatrix}1 & 3\\2 & 4\end{bmatrix}$, $B = \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$, $C = \begin{bmatrix}4 & 1\\1 & 5\end{bmatrix}$ and $I = \begin{bmatrix}1 & 0\\0 & 1\end{bmatrix}$, find $A(B+C) - 14I$. ICSE 2023, 3 marks

SOLUTION

$$B + C = \begin{bmatrix}5 & 3\\3 & 9\end{bmatrix}$$

$$A(B+C) = \begin{bmatrix}5+9 & 3+27\\10+12 & 6+36\end{bmatrix} = \begin{bmatrix}14 & 30\\22 & 42\end{bmatrix}$$

$$14I = \begin{bmatrix}14 & 0\\0 & 14\end{bmatrix}$$

Answer: $\begin{bmatrix}0 & 30\\22 & 28\end{bmatrix}$

5. If $A = \begin{bmatrix}3 & 0\\5 & 1\end{bmatrix}$ and $B = \begin{bmatrix}-4 & 2\\1 & 0\end{bmatrix}$, find $A^2 - 2AB + B^2$. ICSE 2020, 3 marks

SOLUTION

$$A^2 = \begin{bmatrix}9+0 & 0+0\\15+5 & 0+1\end{bmatrix} = \begin{bmatrix}9 & 0\\20 & 1\end{bmatrix}$$

$$AB = \begin{bmatrix}-12+0 & 6+0\\-20+1 & 10+0\end{bmatrix} = \begin{bmatrix}-12 & 6\\-19 & 10\end{bmatrix}, \text{ so } 2AB = \begin{bmatrix}-24 & 12\\-38 & 20\end{bmatrix}$$

$$B^2 = \begin{bmatrix}16+2 & -8+0\\-4+0 & 2+0\end{bmatrix} = \begin{bmatrix}18 & -8\\-4 & 2\end{bmatrix}$$

$$A^2 - 2AB + B^2 = \begin{bmatrix}9+24+18 & 0-12-8\\20+38-4 & 1-20+2\end{bmatrix}$$

Answer: $\begin{bmatrix}51 & -20\\54 & -17\end{bmatrix}$

6. If $A = \begin{bmatrix}2 & 3\\5 & 7\end{bmatrix}$, $B = \begin{bmatrix}0 & 4\\-1 & 7\end{bmatrix}$ and $C = \begin{bmatrix}1 & 0\\-1 & 4\end{bmatrix}$, find $AC + B^2 - 10C$. ICSE 2018, 3 marks

SOLUTION

$$AC = \begin{bmatrix}2-3 & 0+12\\5-7 & 0+28\end{bmatrix} = \begin{bmatrix}-1 & 12\\-2 & 28\end{bmatrix}$$

$$B^2 = \begin{bmatrix}0-4 & 0+28\\0-7 & -4+49\end{bmatrix} = \begin{bmatrix}-4 & 28\\-7 & 45\end{bmatrix} \text{ and } 10C = \begin{bmatrix}10 & 0\\-10 & 40\end{bmatrix}$$

$$AC + B^2 - 10C = \begin{bmatrix}-1-4-10 & 12+28-0\\-2-7+10 & 28+45-40\end{bmatrix}$$

Answer: $\begin{bmatrix}-15 & 40\\1 & 33\end{bmatrix}$

MathSetu

7. $A = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ -3 & 2 \end{bmatrix}$ and $A^2 - 5B^2 = 5C$. Find the matrix C. ICSE 2017, 4 marks

SOLUTION

$$A^2 = \begin{bmatrix} 1+9 & 3+12 \\ 3+12 & 9+16 \end{bmatrix} = \begin{bmatrix} 10 & 15 \\ 15 & 25 \end{bmatrix}$$

$$B^2 = \begin{bmatrix} 4-3 & -2+2 \\ 6-6 & -3+4 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$5C = A^2 - 5B^2 = \begin{bmatrix} 5 & 15 \\ 15 & 20 \end{bmatrix}$$

Answer: $C = \begin{bmatrix} 1 & 3 \\ 3 & 4 \end{bmatrix}$

2 Solving matrix equations

8. $A = \begin{bmatrix} x & 0 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 4 & 0 \\ y & 1 \end{bmatrix}$ and $C = \begin{bmatrix} 4 & 0 \\ x & 1 \end{bmatrix}$. Find x and y if $AB = C$. ICSE 2024, 4 marks

SOLUTION

$$AB = \begin{bmatrix} 4x+0 & 0+0 \\ 4+y & 0+1 \end{bmatrix} = \begin{bmatrix} 4x & 0 \\ 4+y & 1 \end{bmatrix}$$

Compare with C: $4x = 4$, so $x = 1$. Then $4 + y = x = 1$, so $y = -3$.

Answer: $x = 1, y = -3$

9. If $\begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, find x and y. (a) $1, -2$ (b) $-2, 1$ (c) $1, 2$ (d) $-2, -1$ ICSE 2023, 1 mark

SOLUTION

Multiply out: $\begin{bmatrix} 2x \\ 4y \end{bmatrix} = \begin{bmatrix} 2 \\ -8 \end{bmatrix}$, so $x = 1$ and $y = -2$.

Answer: (a)

10. $A = \begin{bmatrix} x & 3 \\ y & 3 \end{bmatrix}$ and $A^2 = 3I$, where I is the identity matrix of order 2. Find x and y. ICSE 2020, 3 marks

SOLUTION

$$A^2 = \begin{bmatrix} x^2+3y & 3x+9 \\ xy+3y & 3y+9 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$$

$3x + 9 = 0$ gives $x = -3$, and $3y + 9 = 3$ gives $y = -2$.

Check: $x^2 + 3y = 9 - 6 = 3$ and $xy + 3y = 6 - 6 = 0$.

Answer: $x = -3, y = -2$

11. Given $\begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix} M = 6I$, where I is the unit matrix of order 2×2, (a) state the order of M (b) find M.

ICSE 2019, 4 marks

SOLUTION

(a) The product is a 2×2 matrix and the first matrix is 2×2, so M is 2×2.

(b) Let $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$. Then $\begin{bmatrix} 4a+2c & 4b+2d \\ -a+c & -b+d \end{bmatrix} = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}$.

$-a + c = 0$ gives $c = a$, and then $6a = 6$, so $a = 1$ and $c = 1$.

$-b + d = 6$ gives $d = b + 6$, and then $4b + 2b + 12 = 0$, so $b = -2$ and $d = 4$.

Answer: $M = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$

12. Given $B = \begin{bmatrix} 1 & 1 \\ 8 & 3 \end{bmatrix}$, find $X = B^2 - 4B$. Then solve $X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$ for a and b. ICSE 2017, 4 marks

SOLUTION

$B^2 = \begin{bmatrix} 1+8 & 1+3 \\ 8+24 & 8+9 \end{bmatrix} = \begin{bmatrix} 9 & 4 \\ 32 & 17 \end{bmatrix}$ and $4B = \begin{bmatrix} 4 & 4 \\ 32 & 12 \end{bmatrix}$

$X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$

$X \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 5a \\ 5b \end{bmatrix} = \begin{bmatrix} 5 \\ 50 \end{bmatrix}$

Answer: $X = \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$; $a = 1, b = 10$

3 Addition, subtraction and multiplication by a number

13. Simplify $\sin A \begin{bmatrix} \sin A & -\cos A \\ \cos A & \sin A \end{bmatrix} + \cos A \begin{bmatrix} \cos A & \sin A \\ -\sin A & \cos A \end{bmatrix}$. ICSE 2019, 3 marks

SOLUTION

Multiply each matrix by its number, then add the matching entries.

$= \begin{bmatrix} \sin^2 A + \cos^2 A & -\sin A \cos A + \sin A \cos A \\ \sin A \cos A - \sin A \cos A & \sin^2 A + \cos^2 A \end{bmatrix}$

Answer: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

14. Find x and y if $2 \begin{bmatrix} x & 7 \\ 9 & y-5 \end{bmatrix} + \begin{bmatrix} 6 & -7 \\ 4 & 5 \end{bmatrix} = \begin{bmatrix} 10 & 7 \\ 22 & 15 \end{bmatrix}$. ICSE 2018, 3 marks

SOLUTION

Left side $= \begin{bmatrix} 2x+6 & 14-7 \\ 18+4 & 2y-10+5 \end{bmatrix} = \begin{bmatrix} 2x+6 & 7 \\ 22 & 2y-5 \end{bmatrix}$

$2x + 6 = 10$ gives $x = 2$, and $2y - 5 = 15$ gives $y = 10$.

Answer: $x = 2, y = 10$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu